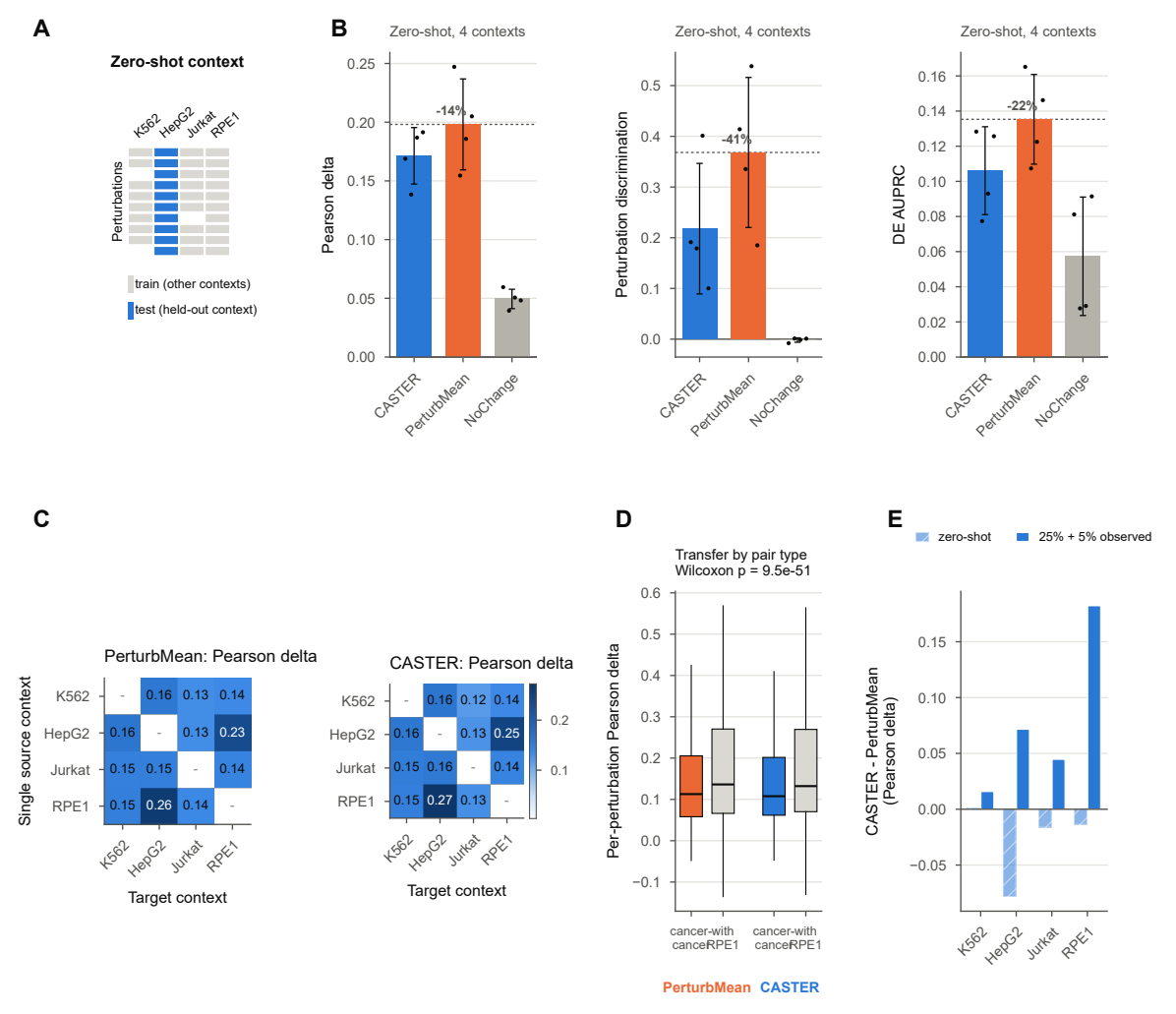

A
Zero-shot context
K562
HepG2
Jurkat
RPE1
Perturbations
train (other contexts)
test (held-out context)
B
Zero-shot, 4 contexts
Pearson delta
-14%
CASTER
PerturbMean
NoChange
Zero-shot, 4 contexts
Perturbation discrimination
-41%
CASTER
PerturbMean
NoChange
Zero-shot, 4 contexts
DE AUPRC
-22%
CASTER
PerturbMean
NoChange
C
PerturbMean: Pearson delta
Single source context
K562
HepG2
Jurkat
RPE1
Target context
CASTER: Pearson delta
Target context
D
Transfer by pair type
Wilcoxon p = 9.5e-51
Per-perturbation Pearson delta
cancer-with cancer
RPE1
PerturbMean
CASTER
E
zero-shot
25% + 5% observed
CASTER - PerturbMean (Pearson delta)
K562
HepG2
Jurkat
RPE1